

HYBRID KINETIC FORMALISM (REMINDER)

Faraday's law

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E},$$

Ampere's law

$$\mu_0 \mathbf{j} = \nabla \times \mathbf{B},$$

Ion Vlasov eq.

$$\frac{\partial f_p}{\partial t} = -\mathbf{v} \cdot \nabla f_p - \frac{\mathbf{E} + \mathbf{v} \times \mathbf{B}}{m_p} \cdot \nabla_{\mathbf{v}} f_p,$$

Ion moments

$$n_i = \sum_p \int f_p(\mathbf{r}, \mathbf{v}) d^3v,$$

$$\mathbf{v}_i = \frac{1}{n_i} \sum_p \int \mathbf{v} f_p(\mathbf{r}, \mathbf{v}) d^3v,$$

Quasi-neutrality

$$n_i = n_e = n,$$

$$\mathbf{v}_e = \mathbf{v}_i - \frac{\mathbf{j}}{ne},$$

Generalized Ohm's law

$$\mathbf{E} = -\mathbf{v}_e \times \mathbf{B} - \frac{1}{en} \nabla \cdot \mathbf{P}_e - \frac{m_e}{e} \frac{d\mathbf{v}_e}{dt}.$$

PARTICLE IN CELL: A STATISTICAL LAGRANGIAN APPROACH

Faraday's law

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E},$$

Ampere's law

$$\mu_0 \mathbf{j} = \nabla \times \mathbf{B},$$

Ion Vlasov eq.

$$\frac{\partial f_p}{\partial t} = -\mathbf{v} \cdot \nabla f_p - \frac{\mathbf{E} + \mathbf{v} \times \mathbf{B}}{m_p} \cdot \nabla_{\mathbf{v}} f_p,$$

Ion moments

$$n_i = \sum_p \int f_p(\mathbf{r}, \mathbf{v}) d^3v,$$

$$\mathbf{v}_i = \frac{1}{n_i} \sum_p \int \mathbf{v} f_p(\mathbf{r}, \mathbf{v}) d^3v,$$

Quasi-neutrality

$$n_i = n_e = n,$$

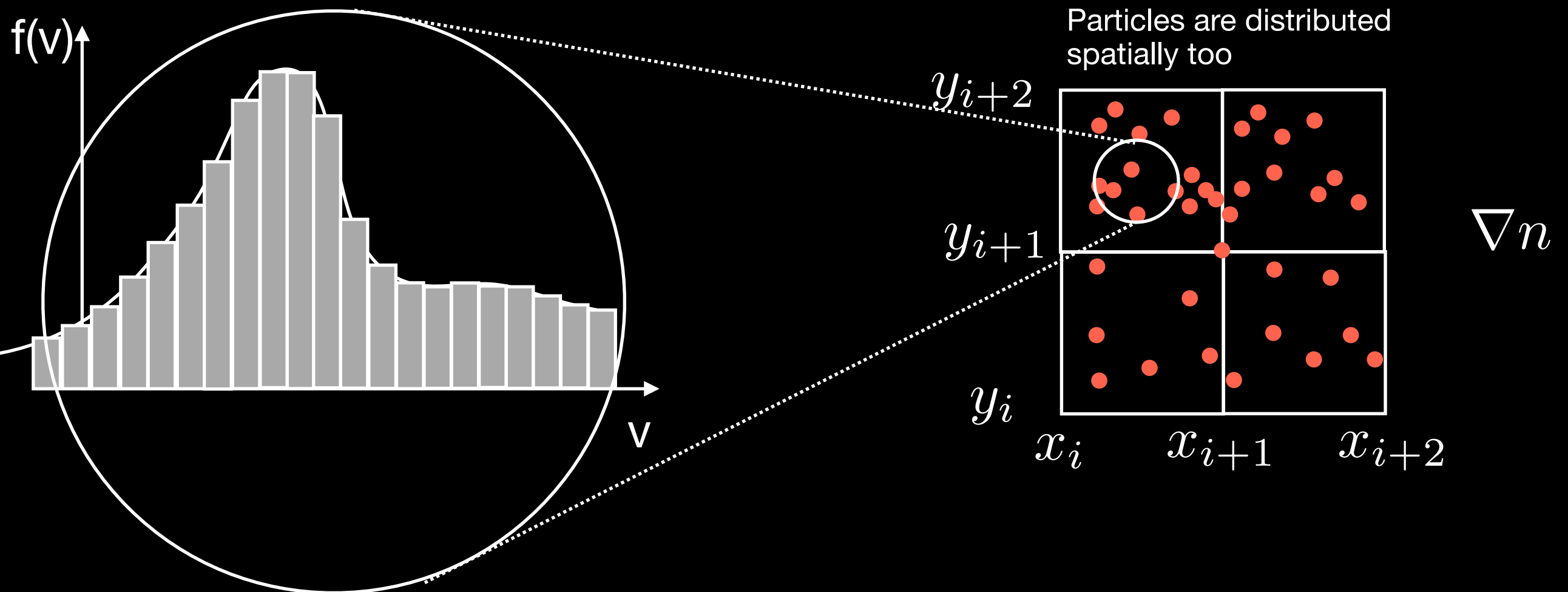
$$\mathbf{v}_e = \mathbf{v}_i - \frac{\mathbf{j}}{ne},$$

Generalized Ohm's law

$$\mathbf{E} = -\mathbf{v}_e \times \mathbf{B} - \frac{1}{en} \nabla \cdot \mathbf{P}_e - \frac{m_e}{e} \frac{d\mathbf{v}_e}{dt}$$

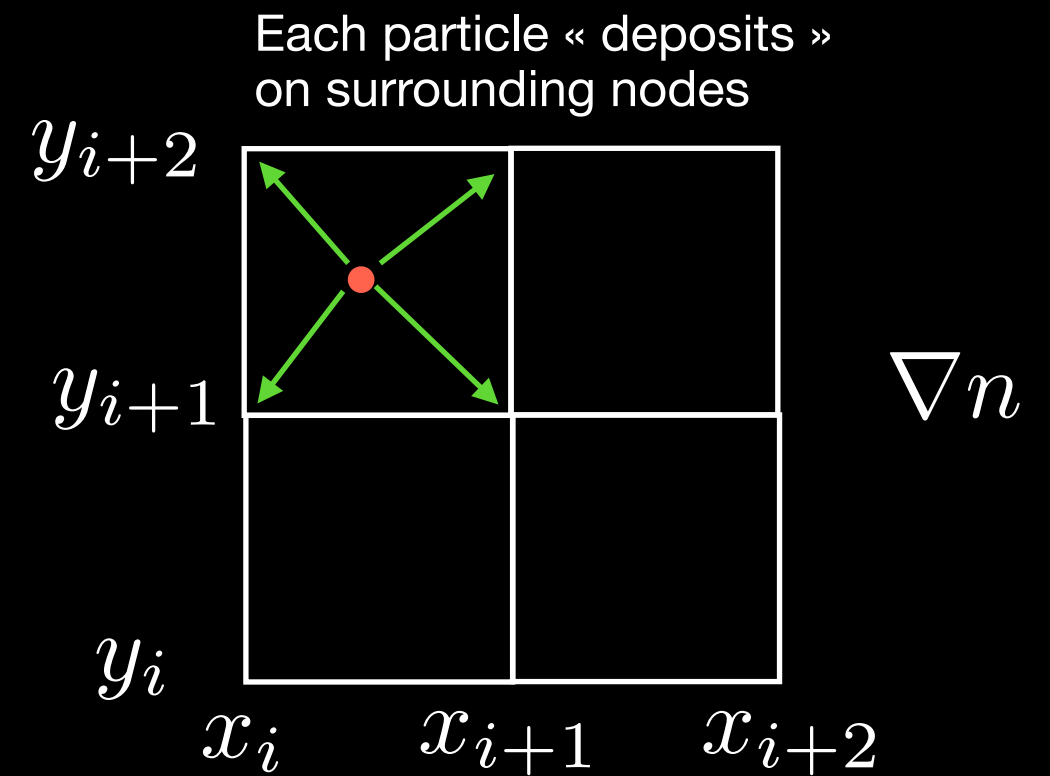
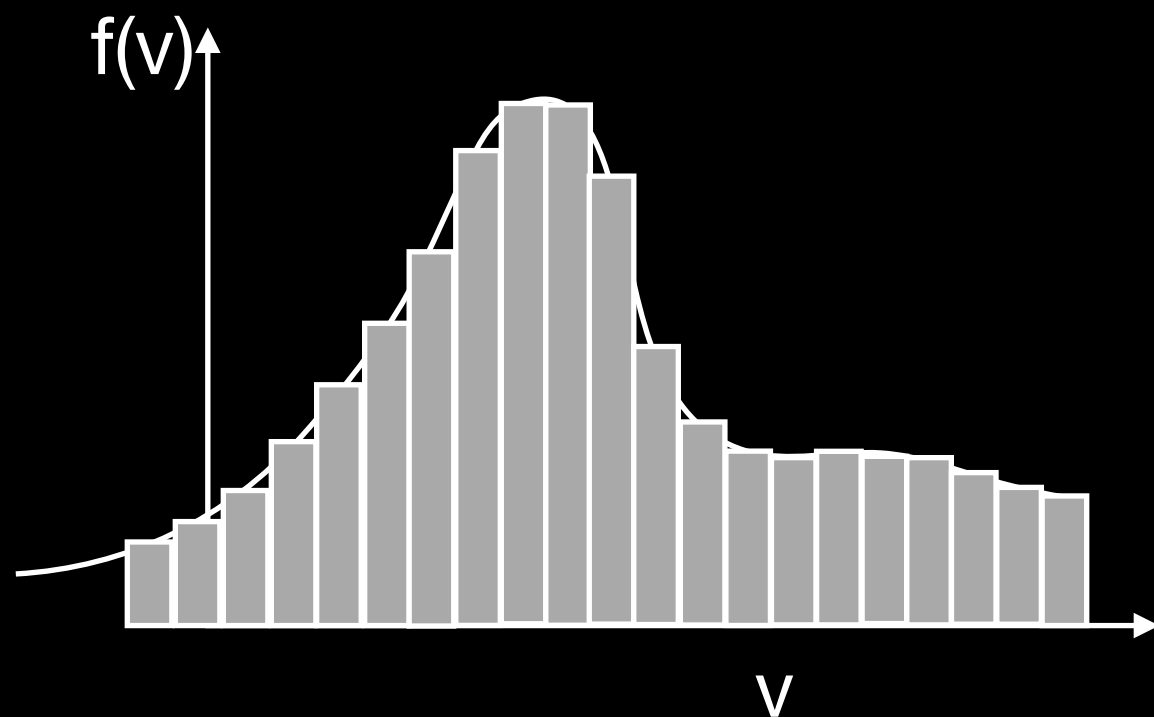
PARTICLE IN CELL: A STATISTICAL LAGRANGIAN APPROACH

$$\frac{\partial f_p}{\partial t} = -\mathbf{v} \cdot \nabla f_p - \frac{\mathbf{E} + \mathbf{v} \times \mathbf{B}}{m_p} \cdot \nabla_{\mathbf{v}} f_p,$$



$f_p(\mathbf{r}, \mathbf{v})$ The distribution is discretely sampled in *phase space*

COMPUTING MOMENTS OF THE DISTRIBUTION FUNCTION ON THE GRID



$$\begin{aligned}
 n_i &= \sum_p \int f_p(\mathbf{r}, \mathbf{v}) d^3v, & \longrightarrow & n_{ij} = \sum_p^N S(\mathbf{r}_p - \mathbf{r}_{ij}) w_p \\
 \mathbf{v}_i &= \frac{1}{n_i} \sum_p \int \mathbf{v} f_p(\mathbf{r}, \mathbf{v}) d^3v, & \longrightarrow & \mathbf{v}_{ij} = \frac{1}{n_{ij}} \sum_p^N \mathbf{v}_p S(\mathbf{r}_p - \mathbf{r}_{ij}) w_p
 \end{aligned}$$

COMPUTING MOMENTS OF THE DISTRIBUTION FUNCTION ON THE GRID

$$n_{ij} = \sum_p^N S(\mathbf{r}_p - \mathbf{r}_{ij}) w_p$$

$$\mathbf{v}_{ij} = \frac{1}{n_{ij}} \sum_p^N \mathbf{v}_p S(\mathbf{r}_p - \mathbf{r}_{ij}) w_p$$

w_p Macroparticle « statistical weight »

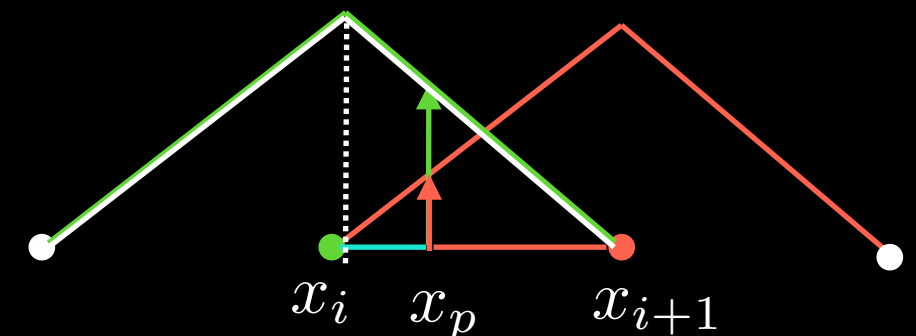
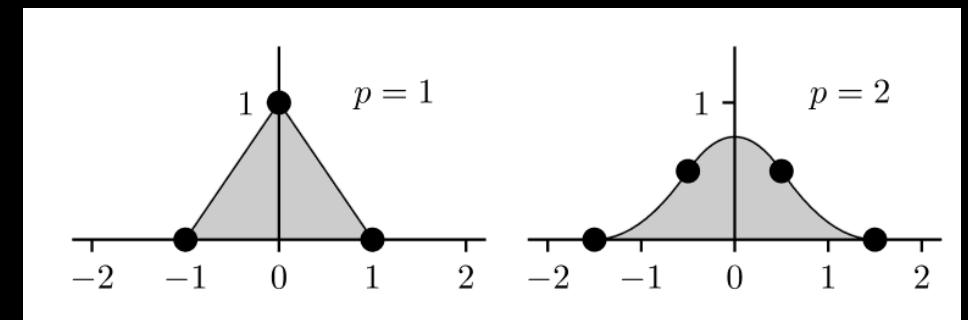
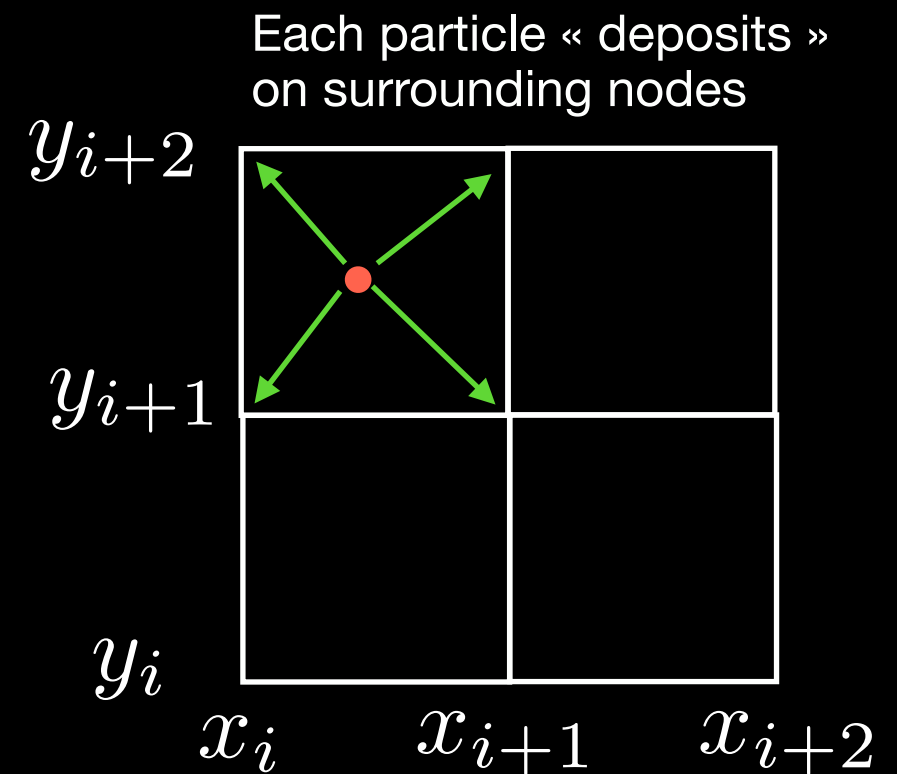
= how much the macro-particle contributes to the local density

Shape function $S(\mathbf{r}_p - \mathbf{r}_{ij})$ B-splines

Order 1 is just a linear interpolation

$$n_{i+1} = \frac{(x_p - x_i)}{\Delta x} w_p$$

$$n_i = \left(1 - \frac{(x_p - x_i)}{\Delta x}\right) w_p$$

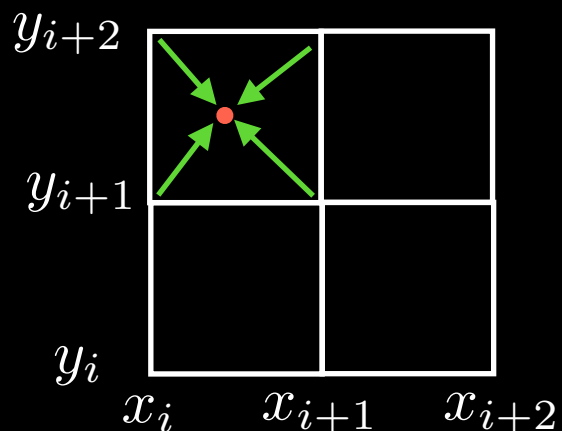


Move particles

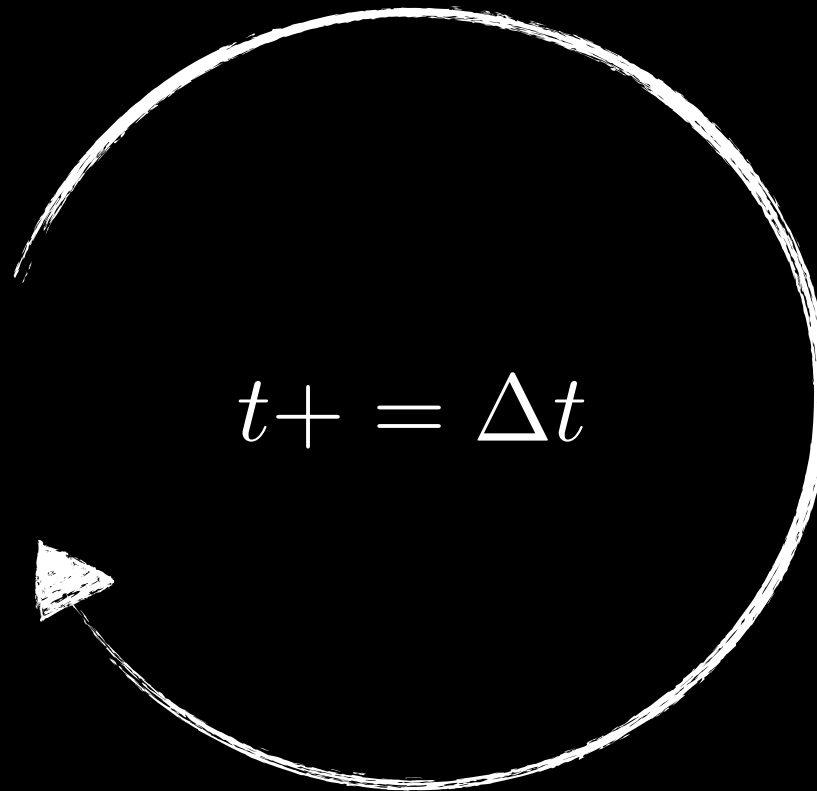
$$m \frac{d\mathbf{v}}{dt} = q (\mathbf{E} + \mathbf{v}_p \times \mathbf{B}) \quad \frac{d\mathbf{r}_p}{dt} = \mathbf{v}_p$$

Fields to particles

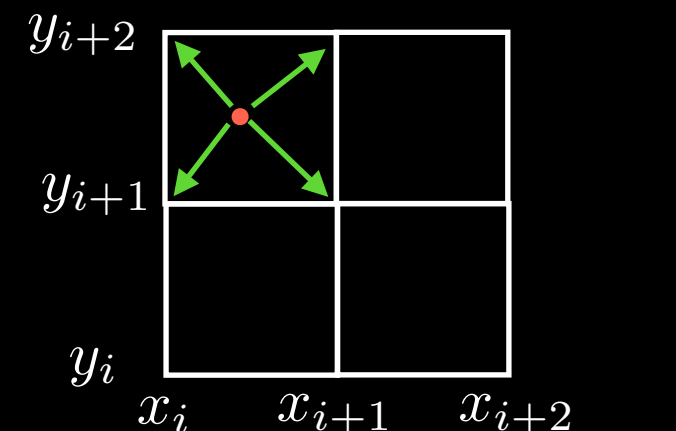
$(\mathbf{E}(\mathbf{r}), \mathbf{B}(\mathbf{r}))$



$$\mathbf{Q}(\mathbf{r}) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} Q_{ij} S(\mathbf{r}_p - \mathbf{r}_{ij})$$



Accumulate moments



$$n_{ij} = \sum_p^N S(\mathbf{r}_p - \mathbf{r}_{ij}) w_p$$

$$\mathbf{v}_{ij} = \frac{1}{n_{ij}} \sum_p^N \mathbf{v}_p S(\mathbf{r}_p - \mathbf{r}_{ij}) w_p$$

Field equations

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}, \quad \mathbf{v}_e = \mathbf{v}_i - \frac{\mathbf{j}}{ne},$$

$$\mu_0 \mathbf{j} = \nabla \times \mathbf{B}, \quad \mathbf{E} = -\mathbf{v}_e \times \mathbf{B} - \frac{1}{en} \nabla \cdot \mathbf{P}_e - \frac{m_e}{e} \frac{d\mathbf{v}_e}{dt}.$$

THE BORIS PUSHER

Mid-step position	$\mathbf{r}^{n+1/2} = \mathbf{r}^n + \mathbf{v}^n \frac{\Delta t}{2}$
Fields on particle	$\mathbf{E}_{ijk}^{n+1/2} \rightarrow \mathbf{E}^{n+1/2}(\mathbf{r}^{n+1/2})$
	$\mathbf{B}_{ijk}^{n+1/2} \rightarrow \mathbf{B}^{n+1/2}(\mathbf{r}^{n+1/2})$
Half electric acceleration	$\mathbf{v}^- = \mathbf{v}^n + \frac{q\Delta t}{2m} \mathbf{E}^{n+1/2}$
Magnetic rotation	$\mathbf{t} = \frac{q\Delta t}{2m} \mathbf{B}^{n+1/2}$
	$\mathbf{v}' = \mathbf{v}^- + \mathbf{v}^- \times \mathbf{t}$
	$\mathbf{s} = \frac{2\mathbf{t}}{1 + t^2}$
	$\mathbf{v}^+ = \mathbf{v}^- + \mathbf{v}' \times \mathbf{s}$
Half electric acceleration	$\mathbf{v}^{n+1} = \mathbf{v}^+ + \frac{q\Delta t}{2m} \mathbf{E}^{n+1/2}$
Final position	$\mathbf{r}^{n+1} = \mathbf{r}^{n+1/2} + \mathbf{v}^{n+1} \frac{\Delta t}{2}$

